

Amortix: amortized Bayesian parameter recovery for dynamical systems via conditional flow matching

Technical report, v0.1

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Abstract

amortix is a small PyTorch package that trains a single conditional-flow-matching (CFM) posterior $p(m \mid \mathcal{D})$ over the parameters m of a dynamical system (ODE, SDE, or PDE), amortized both over datasets and — in its design-amortized mode — over *observation designs*: one network answers $p(m \mid \text{any } K \text{ points } (t_i, x_i))$ for K from a handful to near-continuous monitoring. The emphasis throughout is on universal, learnable components in place of per-problem feature engineering, and on an evaluation protocol whose instrument of record is comparison against exact or validated-MCMC reference posteriors rather than simulation-based calibration (SBC) alone. On a fixed-design gallery of nine systems, 31/32 parameters pass SBC with mean calibration error 1.3pp; on a design-amortized zoo of six further systems (Heston, Merton jump–diffusion, Hénon–Heiles, Hodgkin–Huxley, pharmacokinetics, Fisher–KPP reaction–diffusion), every observed defect either has a learnable architectural cure or lies on a measured compute-budget curve (Merton width excess $\propto 1/\text{budget}$: $1.246 \rightarrow 1.120 \rightarrow 1.059$; Fisher–KPP along-ridge bias $\sim \text{budget}^{-1/2}$: $+0.41 \rightarrow +0.23 \rightarrow +0.14$ posterior-sd). We document three recurring guises of one conditioning disease of set encoders — floating input scale, cardinality blindness, heavy tails — each cured by a learnable module, and record that in three hand-crafted-versus-learned duels the learned counterpart matched the hand fix every time.

1 Scope and method

The method follows amortized simulation-based inference with flow matching [1, 2, 3]. A prior $p(m)$ over box-constrained parameters and a simulator producing observation tokens define the training distribution. Parameters are mapped to $z_1 = \Phi^{-1}(\text{cdf}_{\text{prior}}(m))$ (probit normalization), and a velocity field $v_\theta(z, t, \mathcal{D})$ is trained on the linear path $z_t = (1 - t)z_0 + tz_1$, $z_0 \sim \mathcal{N}(0, I)$, with loss $\|v_\theta - (z_1 - z_0)\|^2$. Sampling integrates the ODE with 20 midpoint steps; an exponential moving average of the weights is used at evaluation.

The conditioning network is a small transformer encoder over observation tokens (default $d = 64$, 3 blocks); the velocity network applies parameter self-attention with adaLN-Zero modulation [7] by the flow time and pooled context (3 blocks). The default model has $\approx 4.8 \times 10^5$ parameters and is deliberately small: capacity was measured *not* to be the accuracy lever (Sec. 5.3).

Two regimes are supported. *Fixed designs*: every dataset shares one observation grid. *Design amortization* (**DesignProblem**): each dataset carries its own set of observation times (and, for PDEs, sensor locations), and the network is trained so that a single model is coherent across all subsets of one measured series — the training mode required for downstream sequential refinement and optimal experimental design.

2 Universal set embeddings and the class rule

The package’s central architectural commitment is that observation tokens are embedded by *learnable*, *problem-agnostic* modules; every hand-crafted feature that ever appeared during development was raced against a learnable counterpart and retired.

Learnable monotone warp. All embeddings share **MonotoneWarp**: a positive mixture over an octave bank of signed-log($1+|x|/s$) scales. It contains the identity and the logarithm in its span, and — critically — in its *initialization basin*: a Yeo–Johnson layer also contains the logarithm (at $\lambda = 0$) but gradient descent never reached it. Hypothesis-class membership is not findability.

Embeddings. For fixed designs: **wdiff/wbasis** (warped values and increments, with a learnable basis mixture). For design amortization the resolution rule is:

- **wfilm** (**SetCondPairEmbed**) when the *observed* process is Markov: tokens are sorted-consecutive-pair features $[t, w(x), \Delta w, \Delta w^2, w_t(\Delta t), \text{first}]$ — exactly the arguments of the Markov likelihood factorization $\prod_i p(x_{i+1} | x_i; m)$ — log-compressed before normalization and modulated by a masked-pool set context via zero-initialized per-channel FiLM.
- **wpoint** (**PointEmbed**) otherwise (hidden state, non-Markov observations): bare points $[t, w(y), \text{meta}, \text{chan}]$.

The class boundary is the Markovianity of the observed process, and it was verified in *both* directions: a pair model transfers cleanly across Markov systems (GBM→OU), while on Heston (hidden volatility) the pair stencil’s advantage disappears — parity, not degradation.

Continuous-time rotary attention. On irregular designs, ordinal RoPE constitutes a hidden train/test contract on token *positions*: training with subsampled tokens left at spread positions while inference packs contiguously produced systematic +1.. + 3 posterior-sd biases with no code being “wrong”. **rope="time"** derives attention phases from physical time with a geometric frequency bank, restoring true permutation invariance (verified to float precision, 10^{-8}).

Cardinality. Softmax attention normalizes — it computes means, and means erase the sample size N that governs posterior width. Encoder forensics (ridge regression from the pooled context onto known sufficient statistics) localized width failures exactly where $R^2(\log N)$ dropped (0.985 for pairs vs. 0.498 for edge-value attention). The cure is metadata, not process knowledge: a log K feature or two learned register tokens; ordinal RoPE’s once-mysterious dense-design advantage is explained by contiguous packing smuggling the count through the maximal position.

3 Canonical training recipe

The recipe that closed the dense-design tail, measured causally on GBM/OU/FHN:

1. **Fresh designs each optimizer step:** every step re-draws each trajectory’s observation subset from its full simulated grid (**fit(retokenize=...)**). Design memorization becomes impossible; one simulation serves unboundedly many designs. Fixed designs with a long budget close the dense tail but overfit into sparse-design overconfidence (width ratio down to 0.86).
2. **Mix- K law:** 50% log-uniform over $[K_{\min}, K_{\max}]$ + 50% uniform over the dense half. Alone this halves the dense excess (GBM 1.27 → 1.14, OU 1.34 → 1.14, FHN dense- K SBC $p=0.000 \rightarrow 0.54$).
3. **Budget scaling:** data and steps are scaled proportionally (constant ≈ 38 epochs).

Result on GBM: sigma width ratio 1.06/1.06/1.05/1.07 at $K = 5/9/20/100$ against the exact per-design posterior.

4 Evaluation protocol: SBC is the screen, exact references are the record

SBC [4] at practical sizes (300–500 simulations) is *provably* blind to width ratios up to ≈ 1.3 , and its per-cell p-values flicker between retrains when residual biases sit at 0.1–0.2 posterior-sd. It also passes a posterior that merely returns the prior. **amortix** therefore uses a two-instrument protocol:

- **Screen:** SBC over mixed designs plus fixed- K buckets (**sbc_design**).

- **Record:** signed centre bias and width ratio against an exact or validated reference posterior on the *same* observed points — closed-form where available (GBM, linear-Gaussian), otherwise adaptive Metropolis with the exact-likelihood factories shipped in the package (Poisson-mixture for Merton, log-normal Bateman for pharmacokinetics, PDE-solve Gaussian for Fisher–KPP). Chains are validated by multi-start agreement before being trusted; the MCMC references used below were checked to worst mean discrepancy 0.099 posterior-sd and width ratios 0.91–1.03 against closed forms where both exist.

Twice during development the two instruments *contradicted* each other, and both times the contradiction exposed a real bug (a padded-mask harness error; an ordinal-RoPE positional contract). Two disagreeing instruments are worth more than either.

A third layer of the protocol is **encoder forensics**: ridge regression from the pooled context onto the known sufficient statistics of the problem separates extraction failures from downstream (flow-side) failures. In the Merton autopsy the failing arm encoded the robust statistics *better* than the healthy arm ($R^2 = 0.967$ on truncated realized variance) — the information was extracted; the failure was a drifting input representation.

5 Results

5.1 Fixed-design gallery (nine systems)

31/32 parameters pass SBC ($p > 0.05$), mean calibration error 1.3pp; the single miss is `stoch_lv` β at $p = 0.04$, at the threshold. Cases: linear-Gaussian (exact-posterior sanity), OU, GBM, CIR, double-well, stochastic Lotka–Volterra, FitzHugh–Nagumo, SEIR, SINDy-SDE. Posterior information extraction (width vs. the exact reference where available) reaches 89–90% on the closed-form cases.

5.2 Design-amortized zoo (six systems)

All on the canonical recipe of Sec. 3; verdicts by the protocol of Sec. 4.

case	dim(m)	embed	verdict
Heston	5	wpoint	all 25 SBC cells pass; pair-vs-point parity confirms the class boundary
Merton jump–diffusion	5	wfilm	25/25 after log-compression + FiLM; residual width on the budget curve (Sec. 5.3)
Hénon–Heiles [8]	3	wpoint	3/3 mixed and every K -bucket down to $K=6$; sparse aliasing yields honest width, not miscalibration
Hodgkin–Huxley	4	wpoint	4/4 mixed; one dense cell (g_K at $K=84$) shows the budget signature
Pharmacokinetics	3	wpoint	3/3 mixed; at $K=6$ blood draws, unbiased ($-0.01..-0.06$ sd) at widths 1.09–1.33 vs. exact-likelihood MCMC
Fisher–KPP (PDE)	2	wpoint	ridge geometry $D \cdot r \approx \text{const}$; along-ridge bias on a budget $^{-1/2}$ curve (Sec. 5.3)

5.3 Budget curves: residuals are a price list, not an open problem

Two independent residuals were driven down by scaling training budget (data and steps together) at fixed capacity:

	×1 budget	×3	×6
Merton width excess (hardest cell)	0.246	0.120	0.059
Fisher–KPP along-ridge bias q (sd)	+0.41	+0.23	+0.14
Fisher–KPP front-speed bias c (sd)	−0.21	−0.14	−0.075

Merton’s width excess decays $\propto 1/\text{budget}$; KPP’s along-ridge bias decays $\sim \text{budget}^{-1/2}$ — the weakly identified direction (weakest CFM gradient signal) converges last, with no wall. Capacity is *not* the lever: doubling width and depth at $\times 3$ budget gave 1.103 vs. 1.120; solver refinement (RK4/60 vs. midpoint/20) gave nothing. At $\times 6$ the hardest statistic of the zoo extracts 89% of the available information, indistinguishable from the closed-form dense cells.

6 Failure anatomy: one disease, three guises

Every hard failure across three zoos reduced to the *conditioning of the set encoder’s input*, in three guises, each cured by a learnable module:

guise	symptom	learnable cure
floating multiplicative scale	GBM σ bias $+0.48$ sd	MonotoneWarp octave mixture
cardinality blindness	widths $1.5\text{--}2.5\times$ at dense K	$\log K$ metadata / register tokens
heavy tails (jumps)	Δw^2 sd 3.1×10^4 ; drifting norm	log1p compression + FiLM

Three duels of a hand fix against a universal learned counterpart (log-coordinate; input cardinality; robust per-dataset scaling vs. learned set-conditioned FiLM: $+0.094 \pm 0.054$ vs. $+0.127 \pm 0.055$, statistically indistinguishable) ended 3:0 for “the learned module matches” — with the same pattern each time: *the hand fix locates the mechanism and the initialization basin; the learnable module placed in that basin is what ships*. The FiLM module needed three correctness conditions, each found the hard way: zero-initialization (start at the working base), normalize-before-modulate (the reverse re-creates the representation drift it cures; measured feature sd 1093), and honest masks (set statistics must never see padding — a bug caught only because SBC and the exact probe disagreed).

7 Baselines

Every claim is read against a three-tier ladder (see `BASELINES.md` in the repository): **Tier 0** — controls: prior-mean (exactly 25% mean absolute error under a uniform prior) and a degree-2 ridge regression on ~ 20 summary statistics; these also unmask prior-returning posteriors that SBC passes. **Tier 1** — classical per-case estimators shipped as `sota()` per problem: exact MLE (OU, GBM), Euler pseudo-MLE (CIR), Kramers–Moyal/SINDy regression, multi-start nonlinear least squares, exact posterior mean (linear-Gaussian). **Tier 2** — reference posteriors (closed-form or validated exact-likelihood MCMC), the instrument of record. **Tier 3** — external context: BayesFlow [6], the `sbi` package, and the `sbibm` benchmark [5], with which the Hodgkin–Huxley and ecology/epidemiology cases overlap by design. `amortix`’s differentiators are the SDE-first zoo with real numerical solvers, variable-design amortization, and the exact-reference evaluation harness.

8 Limitations and outlook

(i) Residual widths and ridge biases are on measured budget curves; sharpening is a compute price list. (ii) SBC power limits mean small residuals require exact references; systems with no tractable likelihood inherit the weaker screen. (iii) Optimal experimental design is the natural next step — the fresh-design training mode already makes $p(m | S)$ coherent across subsets S of one measured series — but is deliberately deferred. (iv) Real-data studies with posterior predictive checks are not yet included.

References

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